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Making Comments Precise and Compact

Group 10

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Keep Comments Compact

Keep the useful information; remove unnecessary wording.

LONGER VERSION

```
// The int is the CategoryType.
// The first float is the score,
// the second is the weight.
```

COMPACT VERSION

```
// CategoryType -> (score, weight)
```

A compact comment can communicate the same structure in one line.

Avoid Ambiguous Pronouns

Make it obvious what “it” or “this” refers to.

UNCLEAR

```
// Insert the data into the cache,
// but check if it's too big first.
```

CLEARER

```
// If the data is small enough,
// insert it into the cache.
```

- Pronouns may refer to more than one thing.
- Name the object or rewrite the sentence.

Goal: readers should not have to inspect the code to decode the comment.

Polish Sloppy Sentences

Prefer direct wording that reveals the actual rule.

WORDY

```
# Depending on whether we've already crawled
# this URL before, give it a different priority.
```

DIRECT

```
# Give higher priority to URLs we've
# never crawled before.
```

- Remove filler and indirect phrasing.
- Keep the important rule explicit.

Compact writing is valuable when it also makes the meaning more precise.

Describe Function Behavior Precisely

Describe what the function actually does.

VAGUE COMMENT

```
// Return the number of lines
// in this file.
```

PRECISE COMMENT

```
// Count how many newline bytes
// ('\n') are in the file.
```

Questions to consider

Empty file? Final newline? Carriage returns? State the behavior that matters.

Use Input/Output Examples That Illustrate Corner Cases

Pick examples that answer likely questions.

EXAMPLE FROM THE CHAPTER

Strip("abba/a/ba", "ab")

returns "/a/"

WHY THIS EXAMPLE?

- Shows repeated characters.
- Clarifies what gets stripped.

A useful example demonstrates behavior that a simple case might hide.

State the Intent of Your Code

Explain the goal, not only the mechanics.

MECHANICS

```
// Iterate through the list in reverse order
```

INTENT

```
// Display each price, from highest to lowest
```

- Intent comments describe the desired result.
- They can help readers notice code that contradicts the goal.

Ask: what is this code trying to achieve?

“Named Function Parameter” Comments

Clarify mysterious values passed to a function.

HARD TO INTERPRET

```
Connect(10, false);
```

PARAMETER NAMES ADDED

```
Connect(/* timeout_ms = */ 10,
        /* use_encryption = */ false);
```

- Useful when arguments are not self-explanatory.
- Especially helpful for Boolean values.

Use selectively—most ordinary function calls do not need these comments.

Use Information-Dense Words

Use precise terms to express familiar programming ideas.

LONG EXPLANATION

Store database data locally so future reads can return it faster.

COMPACT TERM

// This class acts as a caching layer to the database.

Other examples from the chapter

- **heuristic**
- **brute-force**
- **naive solution**
- **canonicalize**